

OCEANSIDE, CA, USA

WMO: 722934

Lat: 33.219N

Lon: 117.349W

Elev: 28

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 53121

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	34.6	36.8	17.7	13.4	55.9	22.5	16.9	54.6	16.9	58.2	15.1	59.4	1.9	30	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	85.7	64.5	82.2	65.8	79.5	66.3	71.3	79.6	70.2	78.0	69.1	76.4	8.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	104.2	76.0	67.0	99.5	75.1	66.0	96.1	74.2	35.1	79.5	34.1	78.4	33.2	76.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.5	11.3	DB	30.1	95.0	2.5	5.0	28.4	98.6	26.9	101.5	25.5	104.3	23.8	107.9	(4)
(5)			WB	27.7	73.5	2.9	1.5	25.6	74.6	23.9	75.5	22.3	76.4	20.2	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	53.8	54.1	56.5	58.7	62.4	65.6	69.6	69.9	68.4	63.7	58.0	53.0		
	DBStd	7.14	4.49	4.40	3.97	3.73	3.23	2.50	2.99	2.96	4.29	4.09	4.34	4.14		
	HDD50	43	14	11	1	0	0	0	0	0	0	0	1	16		
	HDD65	1899	349	307	267	192	91	20	2	1	11	73	215	371		
	CDD50	4125	130	125	202	262	385	468	607	618	551	426	241	110		
	CDD65	506	0	1	2	4	11	39	143	155	112	34	5	0		
	CDH74	2680	44	49	70	122	135	76	413	551	622	419	166	13		
	CDH80	577	6	7	16	38	52	12	39	67	160	134	44	2		
(14) Wind	WSAvg	4.2	3.6	4.2	4.2	4.7	4.9	4.7	4.5	4.3	4.2	3.8	3.5	3.5		
(15) Precipitation	PrecAvg	11.3	2.3	2.3	1.6	0.9	0.3	0.1	0.1	0.0	0.2	0.6	0.9	1.7		
	PrecMax	23.0	9.9	10.0	6.3	2.1	1.1	0.8	1.7	0.4	1.1	5.0	3.0	8.0		
	PrecMin	3.5	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	5.4	2.4	2.2	1.5	0.6	0.3	0.2	0.4	0.1	0.2	1.1	0.8	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	80.6	81.5	86.2	86.6	81.1	84.1	85.5	91.0	89.5	85.1	75.0		
		MCWB	56.0	57.5	59.0	60.3	58.6	67.5	70.3	69.6	70.1	63.7	59.0	53.8		
	2%	DB	73.3	73.3	74.2	77.1	76.1	75.5	80.5	81.6	84.0	83.6	78.9	70.6		
		MCWB	54.3	55.1	57.3	58.6	62.6	66.3	70.0	69.9	69.2	61.3	57.5	53.9		
	5%	DB	69.6	68.3	68.9	71.3	71.8	73.3	78.2	79.2	80.5	78.3	73.1	66.7		
		MCWB	54.3	55.0	57.0	59.2	62.3	64.7	69.1	69.1	68.9	62.2	56.4	54.8		
	10%	DB	64.6	64.1	65.5	67.8	69.5	72.0	76.1	77.0	77.1	74.2	69.6	63.8		
		MCWB	53.9	55.1	57.3	58.7	60.9	63.8	67.9	68.0	67.3	63.5	58.0	54.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	61.6	62.5	65.0	66.2	68.6	72.7	72.6	73.2	70.3	64.0	62.0		
		MCDB	67.8	70.2	71.9	77.6	75.5	77.8	79.7	81.4	84.4	77.5	70.0	66.4		
	2%	WB	59.2	59.2	60.4	62.0	64.1	66.9	70.9	70.8	71.3	67.9	62.5	59.7		
		MCDB	65.1	65.4	67.7	70.6	71.8	74.8	78.0	79.1	80.7	75.9	68.6	63.9		
	5%	WB	57.7	57.9	59.2	60.3	62.6	65.7	69.7	69.6	69.8	66.0	61.2	58.2		
		MCDB	63.7	64.4	65.9	67.9	69.4	72.7	76.8	77.5	77.7	73.2	68.3	62.7		
	10%	WB	56.3	56.6	58.1	59.1	61.5	64.5	68.7	68.7	68.4	64.5	59.7	56.7		
		MCDB	62.5	63.0	64.0	66.2	67.6	70.4	75.3	76.0	75.4	70.9	67.0	61.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	21.4	18.7	18.4	14.6	12.7	12.9	13.8	16.8	20.7	24.2	23.6		
		MCDBR	31.8	30.9	27.6	29.0	21.1	15.9	15.5	17.2	22.0	32.2	33.0	29.4		
		MCWBR	17.8	17.5	15.7	15.6	11.1	8.3	7.4	8.4	10.8	17.4	18.3	17.9		
	5% WB	MCDBR	22.7	22.6	19.6	21.1	17.1	15.7	14.7	15.7	18.9	22.5	24.2	20.5		
		MCWBR	14.6	14.3	12.3	12.5	9.4	8.3	7.1	7.9	9.7	13.7	15.6	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.320	0.338	0.352	0.371	0.368	0.396	0.391	0.362	0.357	0.317	0.319		
	taud		2.568	2.553	2.485	2.452	2.426	2.440	2.434	2.430	2.498	2.482	2.580	2.573		
	Ebn at Noon		290	297	298	297	290	289	281	281	286	279	285	279		
	Edh at Noon		25	29	33	36	37	36	36	36	32	30	25	24		
(44) All-Sky Solar Radiation	RadAvg		1033	1309	1733	2090	2214	2336	2314	2216	1889	1450	1132	935		
	RadStd		66	121	92	111	154	147	128	67	82	102	70	68		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.01	N/A	N/A	+2.77	N/A	N/A	N/A	-218	+385	+198
Regional (47 neighbors)	+1.10	N/A	-3.01	+1.19	N/A	N/A	N/A	-185	+391	+202

Nomenclature: See separate page